Venkata Lakshmi Parimala Pasupuleti

Senior Software Engineer · Production RAG & LLM Systems · Evaluation Frameworks · FastAPI · React · Data Pipelines · Full-Stack AI · Python · GCP/AWS

venkatalakshmiparimalap@gmail.com | +1-703-261-5410 | linkedin.com/in/venkata-lakshmi-parimala-pasupuleti | AgentProbe (open-source) | Virginia · Open to Bay Area

PROFESSIONAL SUMMARY

I build **production RAG and LLM systems** end-to-end — **evaluation frameworks, data pipelines, full-stack delivery** — with strong **architectural judgment** across **Python backend** (FastAPI, Django) and **React/TypeScript frontend**. 9+ years engineering (7+ years at Entrata in production); currently at Intuit integrating Anthropic Claude into financial apps: **production LLM systems, not prototypes** — RAG grounding on customer financial records, LLM-as-judge eval CI (caught 3 prompt regressions pre-release, zero hallucination incidents post-deploy). Two open-source AI systems: **AgentProbe** — **evaluation framework for production AI/ML systems** across 5 LLM providers with 15-endpoint **FastAPI** backend; **Leasel** — **production RAG + agentic system** for complex **data pipelines**, 8-agent LangGraph, 93.6% precision across 1,240-case eval harness. Enterprise data scale: 1M+ property records, 180K regulatory records/year, **ETL pipelines integrating enterprise data sources** at Entrata. Cloud: **AWS** (production), **GCP** (Vertex AI coursework), Azure. M.S. Data Analytics Engineering, George Mason University (GPA 3.87, May 2026). — github.com/venkatalakshmiparimala/AgentProbe

CORE COMPETENCIES

Production RAG Systems | LLM Production Systems | Evaluation Frameworks | FastAPI / Django | React / TypeScript | Data Pipelines | Enterprise Data Integration | Full-Stack AI Delivery | Architectural Judgment | LLM-as-Judge Eval CI | Multi-Agent Architectures | GCP / AWS / Azure

EXPERIENCE

Intuit

Jun 2026 – Present · Virginia, USA

Software Engineer — Production LLM Systems · Financial Services RAG · FastAPI · React · AWS

- Build **production LLM systems, not prototypes**: integrate Anthropic Claude into **FastAPI** services for financial data summaries and natural-language queries; RAG grounding on customer financial records ensures every answer traces to source data with full citation traceability.
- Design and ship **React/TypeScript** frontends on AWS end-to-end; develop backend **data pipelines** with validation and error handling; containerize on Docker and ship via GitHub Actions CI/CD with Prometheus + Grafana observability.
- Build **evaluation framework** for LLM outputs: LLM-as-judge scoring with golden datasets and regression detection; caught 3 prompt regressions before release; zero hallucination incidents post-deployment — measures whether the AI system actually improves outcomes.
- Add guardrails around LLM calls: input validation, sensitive-data handling, fallback paths on model failure; build ML pipelines with scikit-learn and XGBoost from data ingestion through prediction APIs.

Entrata (Apartment & Affordable Housing SaaS)

Oct 2020 – Jul 2024 · 100K+ MAU, 1M+ property records

Senior Software Engineer — Python · Django · Java · React · PostgreSQL · AWS · Full-Stack

- Shipped 20+ features **full-stack** (Python, **Django**, Java/Spring Boot, **React**/TypeScript, PostgreSQL) across 120+ Agile sprints; reviewed 300+ PRs; mentored junior engineers.
- Designed and maintained HUD income-limit ETL: **data pipeline integrating enterprise data sources** at scale — federal source rows → 180K regulatory domain records with validation, rollback, compliance enforcement; 7 annual cycles, zero violations.
- Built RabbitMQ async processing system for bulk tenant record updates with retry and idempotency; optimized 200+ SQL queries 30% via execution plan analysis; Python/Bash automation eliminating 80% of manual operational work.
- AWS S3 PHI migration with Libsodium encryption (100% PHI compliance, zero security incidents); Docker microservices redesign cutting execution time 50%; **architectural judgment** across distributed systems at 100K+ MAU scale.

Entrata India (Db Xento Systems)

Sep 2018 – Oct 2020 · Pune, India

Junior Software Engineer — Python · PHP · Java · PostgreSQL · AWS

- Built property management modules and compliance data pipelines processing 1.6M HUD/Tax Credit records; led AWS S3 cloud migration from legacy server storage; created 56 state-specific housing certification documents.

George Mason University — Research Associate / Teaching Assistant

Aug 2024 – May 2026

AI/ML Research · Fine-Tuning · PyTorch · HuggingFace · Statistical Modeling · GCP Vertex AI

- Fine-tuned classification model (HuggingFace Transformers + PyTorch): full training loop from data prep through fine-tuning, eval, and deploy on Quart, presented to GMU faculty panel. Built predictive models ($R^2 = 0.977$); applied Isolation Forest anomaly detection. **GCP Vertex AI** coursework. Mentored 600+ students.

PORTFOLIO PROJECTS

AgentProbe — Evaluation Framework for Production AI/ML Systems

Evaluation Framework · Production RAG · FastAPI

2025 · Python, FastAPI (15 endpoints), LangGraph (3 agents), ChromaDB, Pinecone, Anthropic, OpenAI, Azure OpenAI, Groq, Ollama · github.com/venkatalakshmiparimala/AgentProbe

- Built a **production evaluation framework for AI/ML systems**: 15-endpoint **FastAPI** backend routing prompts across 5 LLM providers (Anthropic, OpenAI, Azure OpenAI, Groq, Ollama) with LLM Router for dynamic provider selection based on latency, cost, and capability signals.
- **5-metric LLM-as-judge eval engine** (factual accuracy, hallucination rate, retrieval quality, completeness, compliance); CI-integrated regression detection blocks deploys when quality drops; **RAG pipeline** evaluation with ChromaDB + Pinecone; used to evaluate Leasel across 1,240+ test cases — measures whether AI systems actually improve.

Leasel — Production RAG + Agentic System for Complex Data Pipelines

Production RAG · Data Pipeline · LangGraph

2026 · Python, LangGraph (8-agent StateGraph), FastAPI, hybrid RAG (TF-IDF + semantic reranking), SQLite, Anthropic/OpenAI APIs · github.com/venkatalakshmiparimala/Lease_complaince_rag

- **Production RAG + agentic system** with complex **data pipelines**: 8-agent LangGraph StateGraph (triage → retrieval → parallel Config/History/StructuredData agents → reasoning → deterministic validator → HITL escalation); integrates **enterprise data sources** (OCR ingestion, section-aware chunking, live HUD income-limit API).
- Hybrid TF-IDF + semantic reranking **RAG pipeline**: 100% citation correctness, 0% unsupported-claim rate; **evaluation framework** via LLM-as-judge CI harness achieving 93.6% precision, 87.9% F1 across 1,240 test cases — proves the AI system actually improves on baseline.

AI on FHIR — Clinical NLP & LLM API Integration

Full-Stack AI · React + TypeScript

2025 · Python, Flask, React, TypeScript, Docker, BERT, spaCy, LLMs, FHIR REST APIs · github.com/venkatalakshmiparimala/AI_on_FHIR

- **Full-stack** NLP pipeline (Python Flask backend + **React/TypeScript** frontend): translates English/Spanish clinical questions into FHIR API calls via LLM + embeddings; HIPAA guardrails with RBAC; ~60% reduction in manual lookup time; containerized with Docker.

EDUCATION

M.S. Data Analytics Engineering — George Mason University, Virginia — May 2026 · GPA 3.87

B.Tech Computer Science & Engineering — JNTUA, India — 2013–2017

TECHNICAL SKILLS

AI & LLM Systems: Production RAG Systems, LLM Production Systems, Evaluation Frameworks, LLM-as-Judge, Hallucination Detection, Multi-Agent Architectures, LangGraph StateGraph, LangChain, HITL Escalation, Guardrails, Structured Outputs, Prompt Engineering, Anthropic Claude, OpenAI, Azure OpenAI, Groq, Ollama, Embeddings, Vector Databases (ChromaDB, Pinecone, Weaviate, Elasticsearch), Hybrid Retrieval

Python & Backend: Python (9+ yrs), FastAPI, Django, Flask, Java, Spring Boot, Node.js, SQL, PL/SQL, Bash, PostgreSQL, MySQL, Oracle, Redis, MongoDB, SQLite, Snowflake, ETL Pipelines, Data Modeling, Query Optimization, RabbitMQ, Kafka, Celery, scikit-learn, XGBoost, PyTorch, PySpark, BERT, spaCy, HuggingFace

Full-Stack & Frontend: React, TypeScript, JavaScript, Redux, GraphQL, PHP, Laravel, C#, .NET, Kotlin, Scala, REST APIs, Microservices, OAuth

Cloud, DevOps & Observability: AWS (S3, EC2, RDS, Lambda, Glue, SQS, API Gateway, IAM), GCP (Vertex AI), Azure (OpenAI, Data Factory, Databricks, AKS), Docker, Kubernetes, Terraform, GitHub Actions CI/CD, OpenTelemetry, Prometheus, Grafana